

Ditrim Oral Suspension

Oral Solution
(FOR VETERINARY USE ONLY)

COMPOSITION

Sulfadiazine Base.....400g/L
Trimethoprim.....80g/L

PRODUCT DESCRIPTION

A white to pale yellow suspension, odorless.

PHARMACODYNAMICS

Sulfadiazine is a bacteriostatic antimicrobial product which acts by blocking the biosynthesis of folic acid, transporter of monocarbonated units, indispensable for the biosynthesis of nucleic acids. This action is a consequence of the structural analogy between the molecule of Sulfadiazine and para-aminobenzoic acid (PABA).

The association with trimethoprim provides a synergistic effect, particularly potent, which results in bactericidal action. This synergism is the result of the folic acid biosynthesis chain blockage at 2 different levels; Sulfadiazine at the synthetase dihydropteroate level and trimethoprim at the reductase dihydrofolate level.

Principal action: The two active ingredients produce a sequential double blockage of bacterial synthesis of folinic acid, giving a level of activity many times greater than that obtained from either ingredient alone.

The in vitro activity covers most common Gram-positive and Gram-negative bacteria including: *Actinobacillus* spp., *Actinomyces bovis*, *Bordetella* spp., *Corynebacterium* and *Arcanobacterium* spp., *Escherichia coli*, *Fusobacterium necrophorum*, *Haemophilus* spp., *Klebsiella* spp., *Listeria monocytogenes*, *Nocardia* spp., *Pasteurella* and *Manheimia* spp., *Proteus* spp., *Salmonella* spp., *Staphylococcus* spp., *Streptococcus* spp.

PHARMACOKINETICS

Following oral administration of sulfadiazine and trimethoprim to chickens, $t_{1/2\alpha}$ and $t_{1/2\beta}$ values of 0.756 and 7.07 hours and 0.680 and 6.24 hours were obtained respectively. Values for T_{max} were 2.46 and

2.44 hours, values for C_{max} were 86.45 and 3.45 mcg/ml and values of AUC were 620.50 and 19.87 mcg. Hour/ml, respectively, for sulfadiazine and trimethoprim.

INDICATIONS

Swine: For the treatment and control of diseases sensitive to trimethoprim/ sulfadiazine in pigs, in particular the respiratory disease complex.
Chickens: For the treatment and control of diseases sensitive to trimethoprim/ sulfadiazine in chickens, in particular enteric and respiratory bacterial infections.

CONTRA-INDICATION

Do not administer in chickens producing eggs for human consumption.

Do not administer to animals with known sulphonamide hypersensitivity.

Contraindication: Administration to unweaned piglets less than 15 to 20 days old. Administration to animals with parenchymal hepatic damage or blood dyscrasias.

ADMINISTRATION AND DOSAGE

Recommended dilution rates apply when water consumption is normal; if subnormal, increase concentration to administer 15 to 30mg/kg total liveweight daily (active ingredients). Continue treatment for 3-5 days according to severity of disease.
Chickens: *E. coli*, *Pasteurella*, *Haemophilus* spp., or *Staphylococcal* infections, 1ml per 4.5 liters of drinking water for 3-5 days. *Salmonella* spp. infections: 3ml per 4.5 liters of drinking water for 3-5 days.
PIGS: 3ml/ 4.5L water.

INTERACTION WITH OTHER MEDICINE

Interaction of product with: Amantadine, antacids, digoxin, diuretics, tricyclic antidepressants, warfarin and phenytoin

PREGNANCY AND LACTATION

Data on the exact level below which no effects on foetal development were observed are not available. However, extensive use of the product in different species over many years, have not shown adverse effects on the foetus.

When administered to lactating females, small amounts of trimethoprim and sulfadiazine are present in the maternal milk. Since no studies have ever been reported of the effect on the development new born young of the ingestion milk, it would be prudent not

to feed very young animals with milk obtained from the mother.

In the absence of toxicology study, this veterinary medicinal product must not be used during pregnancy and lactation.

Do not use in laying birds producing eggs for human consumption.

SIDE EFFECT

Hypersensitivity to sulphonamides has been rarely reported, however this remains a potential undesirable effect. Crystalluria caused by precipitation of insoluble sulphonamide is a theoretical risk although rarely reported.

OVERDOSE AND TREATMENT

No information available. As there is no specific antidote, treatment should be symptomatic.

WITHDRAWAL TIME

Meat :14 days

WARNING AND PRECAUTIONS

After mix drug into drinking water should be used within 1 days.

Shake well before use.

Do not administer in chickens producing eggs for human consumption.

SHELF LIFE

Shelf life of the veterinary medicinal product as packaged for sale: 2 years

Shelf life after first opening of the immediate packaging: 24 hours

Shelf life after dilution or reconstitution according to directions: 24 hours

STORAGE

Store in a cool and dry place around 25°C

Do not store above 30°C and keep away from light.

PACKAGING

1L

Jan 2014

DATE OF REVISION